NANYANG TECHNOLOGICAL UNIVERSITY SINGAPORE

MT6102
The Economics of Maritime Regulations





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Agenda

1. Why do we need regulations?

2. How much and how should we regulate?

3. How do firms and markets respond once regulation is imposed?

1. Why do we need regulations?



George Akerlof
Pioneer of Information Asymmetry

By the end of Part I, students should understand that regulation can be economically justified by market failure, particularly externalities, information asymmetry, principal–agent problems and collective-action problems.

1.1 Why regulate Maritime Transport?

Suppose tomorrow all IMO regulations disappeared. Would shipowners voluntarily maintain exactly the same levels of safety, pollution control and crew competency?

Why would a profit-maximising shipowner invest in safety?

1.2 Can Markets Regulate Themselves?

- There are ***private incentives***:
  - Accidents damage ships
  - Insurance premiums may rise
  - Customers may avoid unsafe operations
  - Crew and cargo may be lost or damaged
  - Reputation may suffer

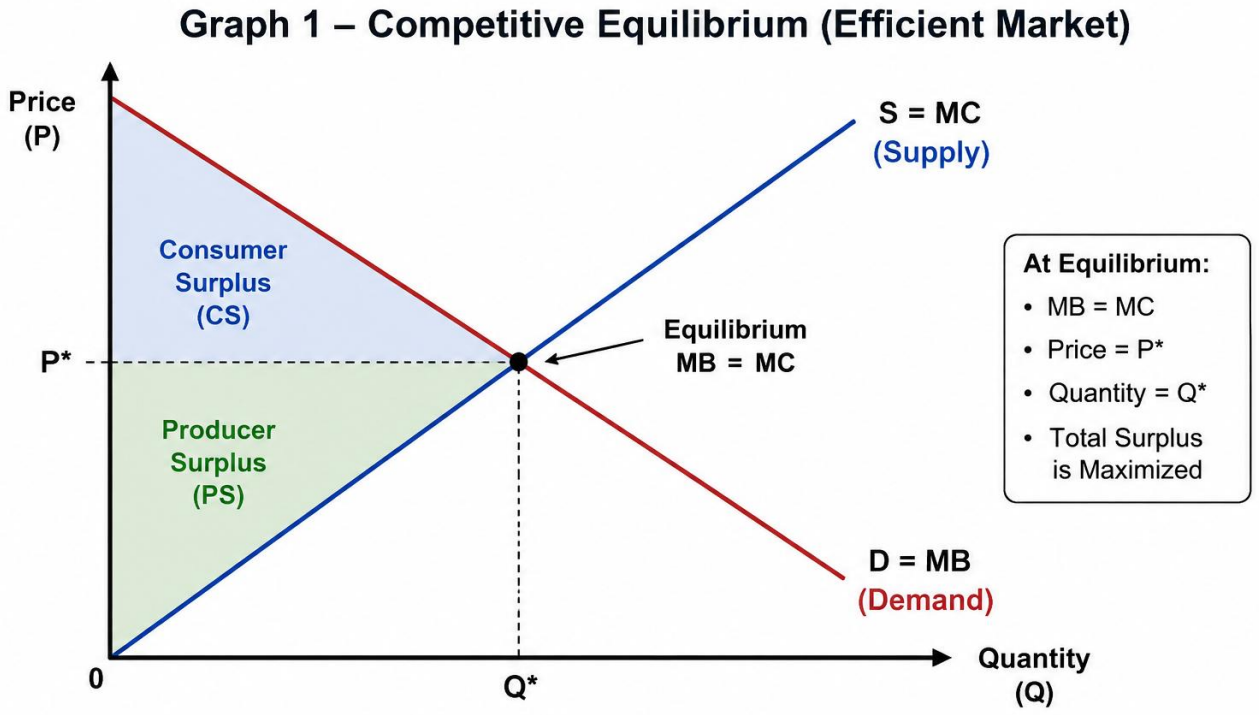
Are private incentives sufficient to produce safety or environmental protection outcomes that are best for the society?

1.3 Efficient Markets: MB = MC

- **Welfare economics**

Under ***ideal*** **competitive-market** conditions, voluntary exchange can produce an efficient allocation of resources.

1.3 Efficient Markets: MB = MC



If all relevant benefits and costs were captured in market transactions, there would be much less economic justification for regulation.

1.4 When Markets Fail

- **Market Failure**
- A situation in which the unregulated market does not **allocate resources efficiently**, resulting in a potential loss of social welfare.

- **Private optimum**

Firm maximises: $\pi = TR - TC_{private}$

- **Social optimum**

Society considers: $Social\ Welfare = Total\ benefits - Total\ Social\ Costs$



1.4 When Markets Fail

Causes of Market Failure

Situations where markets, left on their own, may not produce efficient or desirable outcomes

1 Externalities



Costs or benefits are not fully reflected in private decisions, affecting third parties.

2 Information asymmetry



One party has more or better information than the other, leading to inefficient outcomes.

3 Principal–agent problems



The agent (e.g. manager or crew) may not always act in the best interests of the principal (e.g. shipowner).

4 Collective-action/ free-rider problems



Many actors need to cooperate, but each has an incentive to free ride on the efforts of others, leading to under-provision of public goods (e.g. clean oceans, climate stability).

1.4.1 Negative Externalities

- **What is an externality?**

An externality occurs when the **action** of a producer or consumer **affects another party** and that effect is **not fully reflected** in the market price.

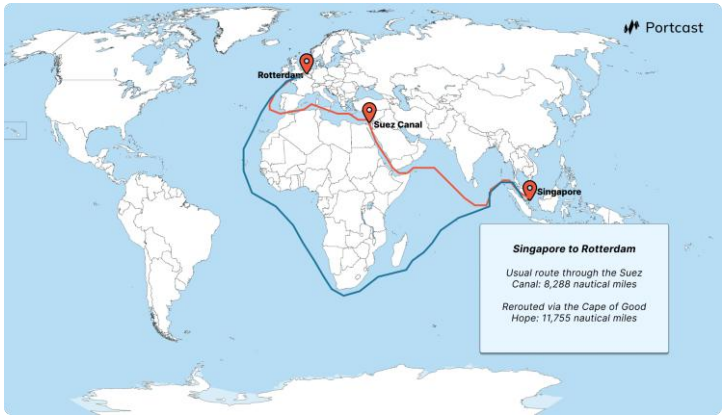
1.4.1 Negative Externalities

- **Take a ship sailing from Singapore to Rotterdam.**

The shipowner considers **1. private costs** which include fuel, crew, maintenance, capital, and port charge.

But its operation can generate:

- Greenhouse emissions (CO_2, SOx & NOx)
- Underwater noise
- Environmental risks



contributing to climate change, diseases, fishery industry, and recreational activities. These **2. external costs** can affect people who are not parties to the shipping transaction.

1.4.1 Negative Externalities

- **Take a ship sailing from Singapore to Rotterdam.**

Therefore:

$$Social\ Cost = Private\ Cost + External\ Cost$$

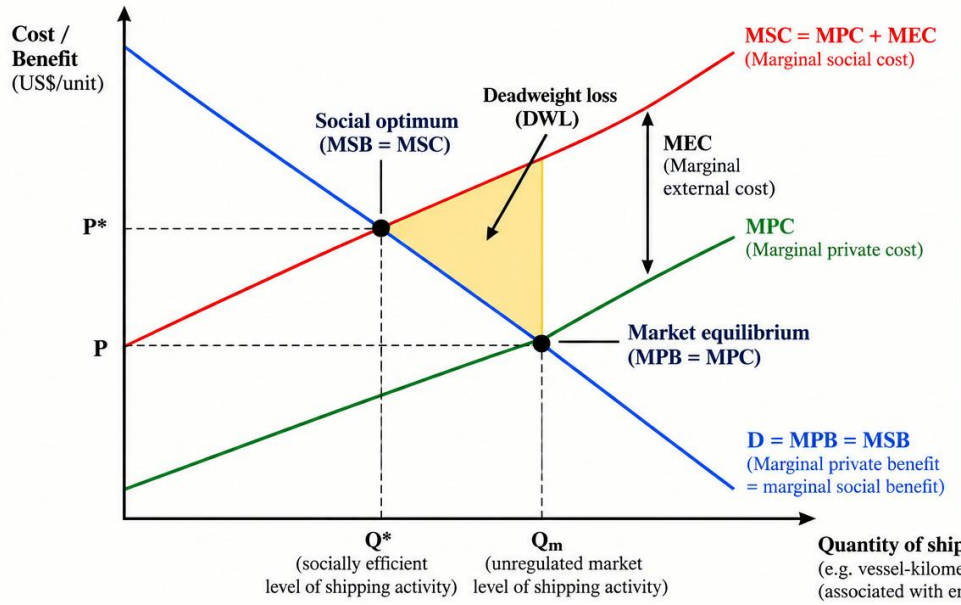
At the margin:

$$Marginal\ Social\ Cost = Marginal\ Private\ Cost + Marginal\ External\ Cost$$

1.4.1 Negative Externalities

Graph 2 – Negative Externality in Maritime Transport

The unregulated market produces too much shipping activity relative to the social optimum



Key points:

- **MPC:** marginal private cost (borne by the shipowner).
- **MEC:** marginal external cost (e.g. climate change, air pollution, health impacts, marine ecosystem damage).
- **MSC = MPC + MEC:** marginal social cost.
- Unregulated market equilibrium at Q_m where MPB = MPC.
- Socially efficient level of shipping activity at Q* where MSB = MSC.
- $Q_m > Q^*$: the market produces too much shipping activity — and therefore too many associated emissions — relative to the social optimum.
- Shaded triangle = *deadweight loss* (welfare loss to society).

Because the shipowner does not bear the full social cost of its operations, the market produces more shipping activity (and associated emissions) than is socially desirable.

1.4.2 Information Asymmetry

- **What is information asymmetry?**

Information asymmetry occurs when one party to an economic transaction has **more or better information** than the other party, potentially leading to **inefficient** market outcomes.

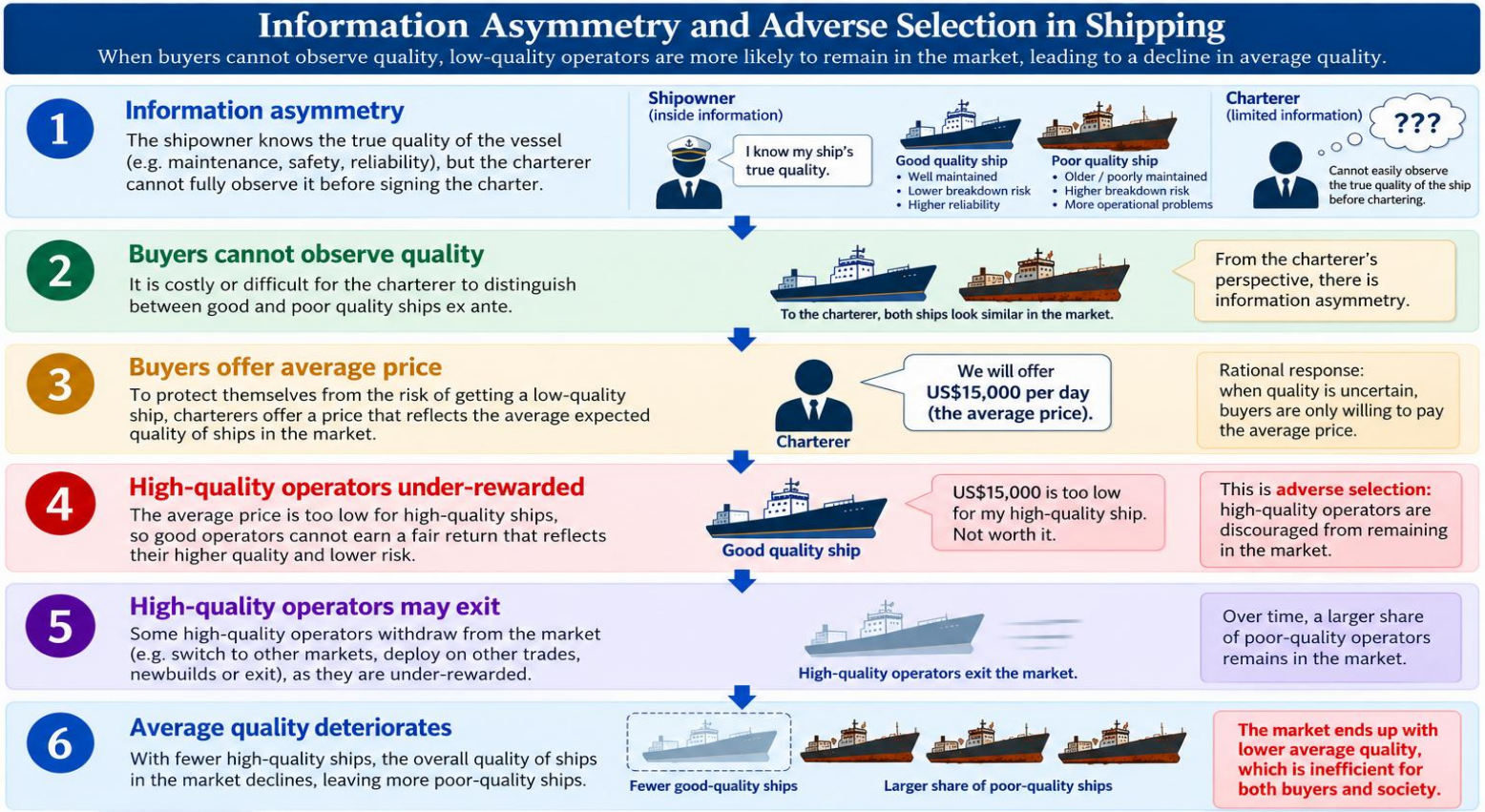
1.4.2 Information Asymmetry

- **What is information asymmetry?**

In maritime transport, the shipowner or operator typically knows more about the **true quality and condition of the vessel** than a charterer, cargo owner, insurer, or other market participant.

$$Information_{Owner} > Information_{Charterer}$$

1.4.2 Information Asymmetry



1.4.2 Information Asymmetry

- **Regulation as an information mechanism**

Regulations through classification societies, statutory certificates, safety audits, port state control, STCW certificates can reduce information asymmetry.

Certifications can act as an observable signal to unobservable quality.

1.4.3 Principal–agent problem

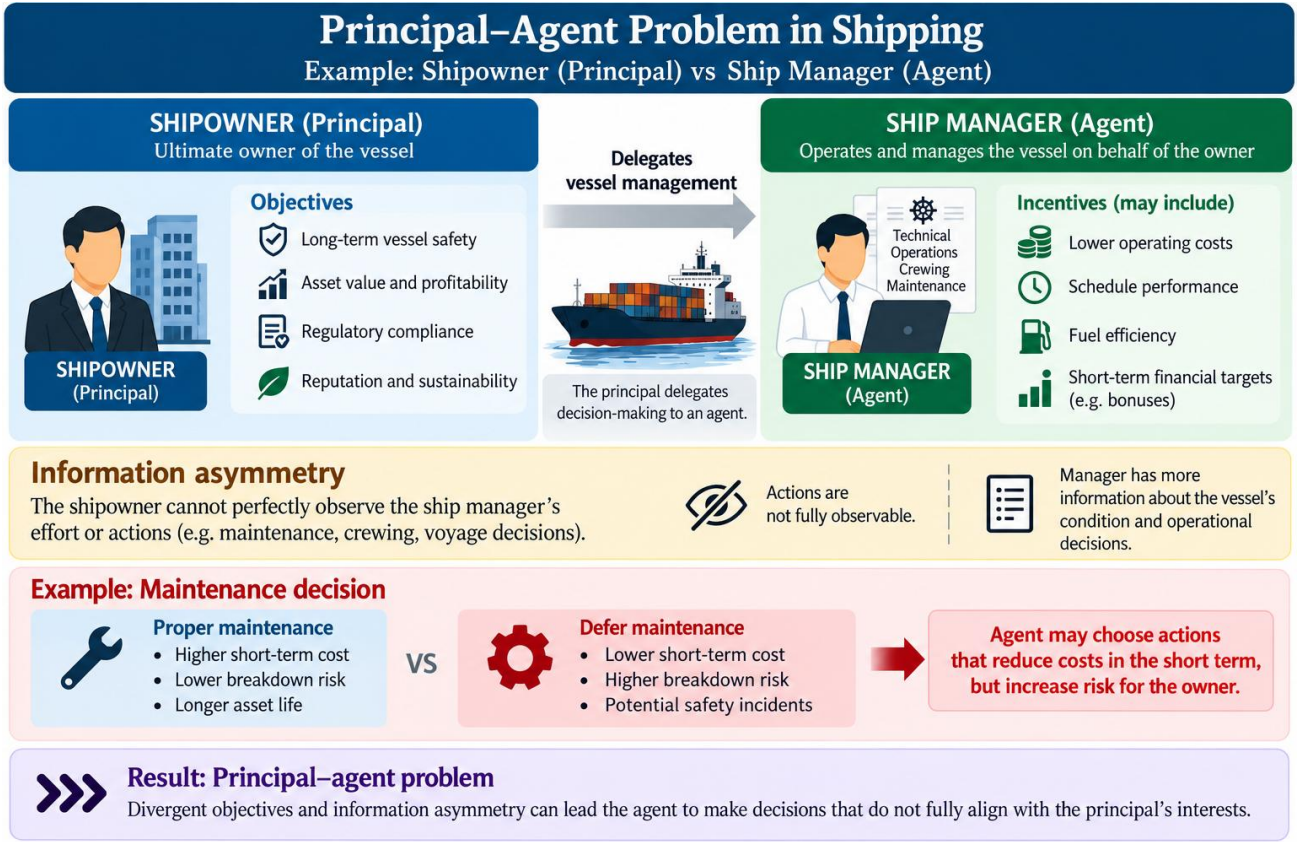
- **Principal–Agent Problem**

A principal–agent problem arises when the principal delegates decisions to an agent whose **interests may differ** and whose **actions cannot be perfectly observed**.

$$Objectives_{Principal} \neq Objectives_{Agent}$$

$$Information_{Principal} < Information_{Agent}$$

1.4.3 Principal–agent problem



1.4.3 Principal–agent problem to Moral Hazard

What happens if the agent's actions cannot be perfectly observed?

- **Moral Hazard**

Moral hazard occurs when a party changes its behaviour or **takes greater risks** because its actions are **imperfectly observable** and it does not bear the **full consequences**.

1.4.3 Principal–agent problem to Moral Hazard

- **Moral Hazard**

Ship manager may defer maintenance, reduce safety expenditure, prioritise schedule over safety, or under-report problems.

1.4.4 Collective action and free-rider problem

- **Collective Action**

A collective action problem occurs when each individual or firm has an incentive not to contribute because the **costs are borne individually** while the **benefits are shared collectively**.

$$\underbrace{Cost\ of\ decarbonisation}_{\text{Borne by individual shipowner}} \quad vs. \quad \underbrace{Climate\ Benefits}_{\text{Shared globally}}$$

1.4.5 Collective action and free-rider problem

- **Collective Action** (Example)

Reducing shipping emissions produces a global benefit, but individual shipowners bear the costs of:

- cleaner but more expensive fuels
- energy-efficient technologies
- vessel retrofits
- investment in low/zero-carbon vessels

1.4.5 Collective action and free-rider problem

- **Free-Rider Problem**

A free rider is an individual or firm that enjoys the benefits of a collective action without contributing to its cost.

1.4.5 Collective action and free-rider problem

- **Free-Rider Problem**

All shipowners benefit from a cleaner environment

↓

Individual owner has an incentive to
FREE RIDE by avoiding emission-reduction costs

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Others bear the cost of reducing emissions

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Too little voluntary emission reduction

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Collective environmental target not achieved

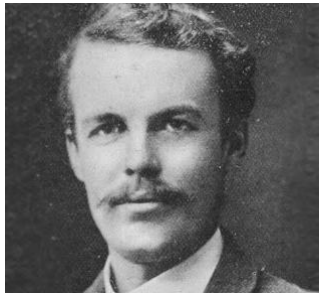
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Collective need for collective regulation

Summary


WHY DOES THE MARKET FAIL?
EXTERNALITIES
INFORMATION ASYMMETRY
AGENCY PROBLEMS
COLLECTIVE ACTION / FREE-RIDER PROBLEMS
Pollution costs not priced into the market
(e.g. emissions, noise, congestion, environmental damage)
Poor quality difficult to observe
(e.g. vessel true condition, safety practices, effort level)
Hidden actions / conflicting incentives
(e.g. effort shirking, manager–owner conflicts, short-term vs long-term objectives)
Benefits are shared, but costs are borne individually
(e.g. firms may free ride on the efforts of others, such as emission reductions or safety investments)
MARKET OUTCOME MAY NOT MAXIMISE SOCIAL WELFARE
POTENTIAL CASE FOR REGULATION

2. How Much and How Should We Regulate?



A. C. Pigou
Pioneer of Externality Taxation

By the end of Part II, students should understand how marginal analysis can determine the optimal level of regulation, and how command-and-control and market-based instruments can be used to internalise externalities and achieve regulatory objectives cost-effectively.

2.1 How much regulation is enough?

If pollution and maritime accidents impose costs on society, why don't we regulate until they are completely eliminated?

2.1 How much regulation is enough?



- **The economic problem**

Benefits of regulation	Costs of regulation
• Fewer accidents and fatalities • Lower emissions and pollution • Reduced environmental damage • Improved health and safety	• Vessel retrofits and new technologies • More expensive fuels • Training and additional manpower • Monitoring and reporting • Administrative and enforcement cost

2.1 How much regulation is enough?

- **The economic problem**

$$More\ regulation \neq Better\ regulation$$

At what point do the additional benefits of tighter regulation no longer justify the additional costs?

2.2 Marginal Analysis of Regulation

- **Marginal Benefit of Regulation (MB)**

The Marginal Benefit (MB) of regulation is the additional social benefit obtained from one additional unit of regulatory stringency.

$$MB_R = \left\{ \begin{array}{c} Additional\ accidents\ prevented \\ Additional\ lives\ saved \\ Additional\ tonnes\ of\ emission\ avoided \\ Addtional\ environmental\ damage\ prevented \end{array} \right\}$$

MB_R typically **reduces** as **regulation increases** because the most valuable or easiest improvements are generally obtained first.

2.2 Marginal Analysis of Regulation

- **Marginal Cost of Regulation (MC)**

The Marginal Cost (MC) of regulation is the additional cost to society arising from one additional unit of regulatory stringency.

$$MC_R = \left\{ \begin{matrix} Increasingly\ expensive\ technologies \\ Higher\ fuel\ costs \\ Additional\ inspections \\ Reporting\ operational\ restrictions \end{matrix} \right\}$$

MC_R typically **increases** as **regulation increases** because the cheapest or easiest compliance measures are generally undertaken first, leaving progressively more costly measures to achieve further improvements.

2.2 Marginal Analysis of Regulation

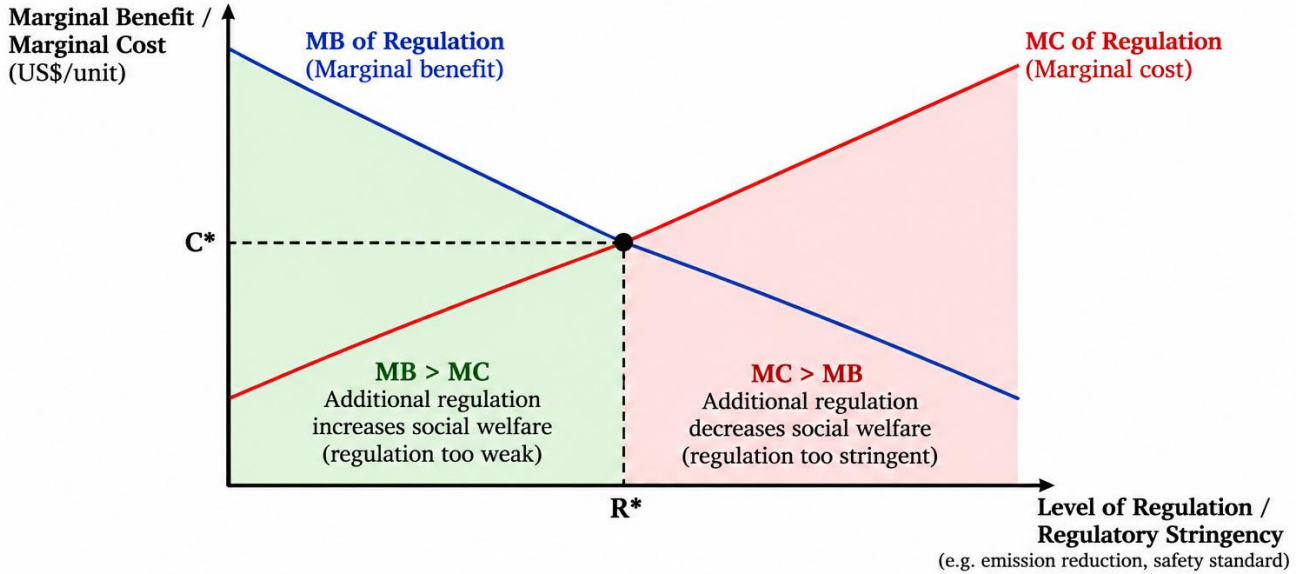
- **Decision Framework**

$$\boldsymbol{Regulate\ further\ when\ MB_R > MC_R}$$

$$\boldsymbol{Do\ not\ regulate\ further\ when\ MC_R > MB_R}$$

2.2 Marginal Analysis of Regulation

Graph 3 – The Economically Optimal Level of Regulation



The economically optimal level of regulation occurs where the marginal benefit of additional regulation equals the marginal cost.

R*: MB = MC

The socially optimal level of pollution or risk is not necessarily zero.

2.2 Marginal Analysis of Regulation



- **Maritime emissions example**

A regulator is considering progressively stricter requirements to reduce emissions from shipping. At approximately what level of emission reduction should the regulator stop tightening the regulation?

Additional emission reduction	Marginal Benefit	Marginal Cost	Net Benefit
First 10%	$10m	$2m	+$8m
Next 10%	$8m	$4m	+$4m
Next 10%	$6m	$6m	$0m
Next 10%	$4m	$8m	−$4m
Final 10%	$2m	$12m	−$10m

2.3 Internalising the Externality

- **What does internalisation mean?**

Internalisation of an externality occurs when the decision-maker is made to take the **external costs or benefits** of its actions into account in its **private decision-making**.

2.3 Internalising the Externality

- **What does internalisation mean?**

The objective is therefore to make the shipowner face a cost closer to:

$$\boldsymbol{Marginal\ Social\ Cost\ (MSC)}$$
$$= Marginal\ Private\ Cost\ (MPC) + Marginal\ External\ Cost\ (MEC)$$

How can a regulator make an external cost become a private cost?

2.3 Internalising the Externality

- **Pigouvian Tax: Putting a Price on Externality**

A tax imposed on an activity generating a negative externality, ideally set equal to the **marginal external cost of that activity.**

At the socially optimal level:

$$Tax = MEC$$

Therefore:

$$MPC + Tax = MSC$$

2.3 Internalising the Externality

- **Pigouvian Tax: Putting a Price on Externality**

Suppose a voyage generates 5,000 *tonnes* of CO_2. A carbon tax of \$100/tCO_2 can be imposed that amounts to 5,000 *tonnes* x \$100 = \$500,000.

The shipowner now has an economic incentive to consider:

- slower steaming
- more efficient vessels
- energy-saving technologies
- alternative fuels
- operational optimisation.

2.4 How should we regulate?

1. Command-and-Control Regulation

Establishes mandatory rules, standards or technologies that regulated firms **must follow**.

- Fuel sulphur limits
- Mandatory safety equipment
- Minimum crew competency requirements
- Technical emission standards
- Mandatory vessel design requirements

2. Market-Based Instrument

Uses prices or market incentives to influence firms' behaviour while allowing firms **flexibility** in how they comply.

- Carbon tax
- Emission trading
- Emission charges

2.4.1 Why does flexibility matter?



- **Maritime tax emissions example**

Suppose two ships each emit 100 tonnes of CO_2. The regulator wants total emissions reduced by 100 tonnes. Suppose command-and-control mandates a limit of 50 tonnes for each ship.

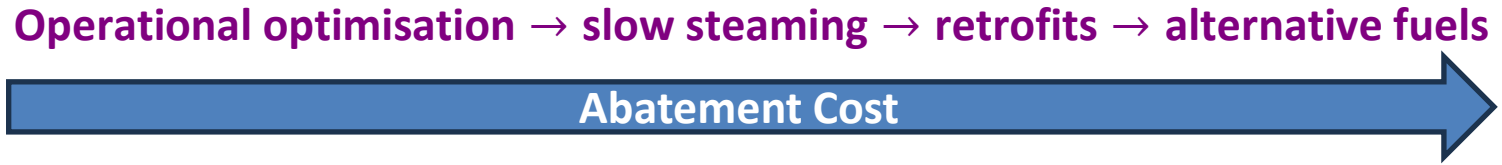
Ship	Description	Abatement Cost
A	Modern vessel with inexpensive efficiency improvements	$\$40/tCO_2$
B	Older vessel requiring expensive modifications	$\$150/tCO_2$

Would it be cheaper for society if Ship A reduced more and Ship B reduced less?

2.4.2 Marginal Abatement Cost (MAC)

- **Marginal Abatement Cost (MAC)**

The **additional cost** of **reducing one additional unit** of pollution or emissions. MAC generally rises as abatement increases as firms adopt the cheapest emission reductions first.



2.4.2 Marginal Abatement Cost (MAC)

- **Marginal Abatement Cost (MAC)**

Suppose:

$$MAC_A = \$40/tCO_2$$
$$MAC_B = \$150/tCO_2$$

and carbon price:

$$P_C = \$100/tCO_2$$

For Ship A:

$$MAC_A < P_C$$

Abate emissions

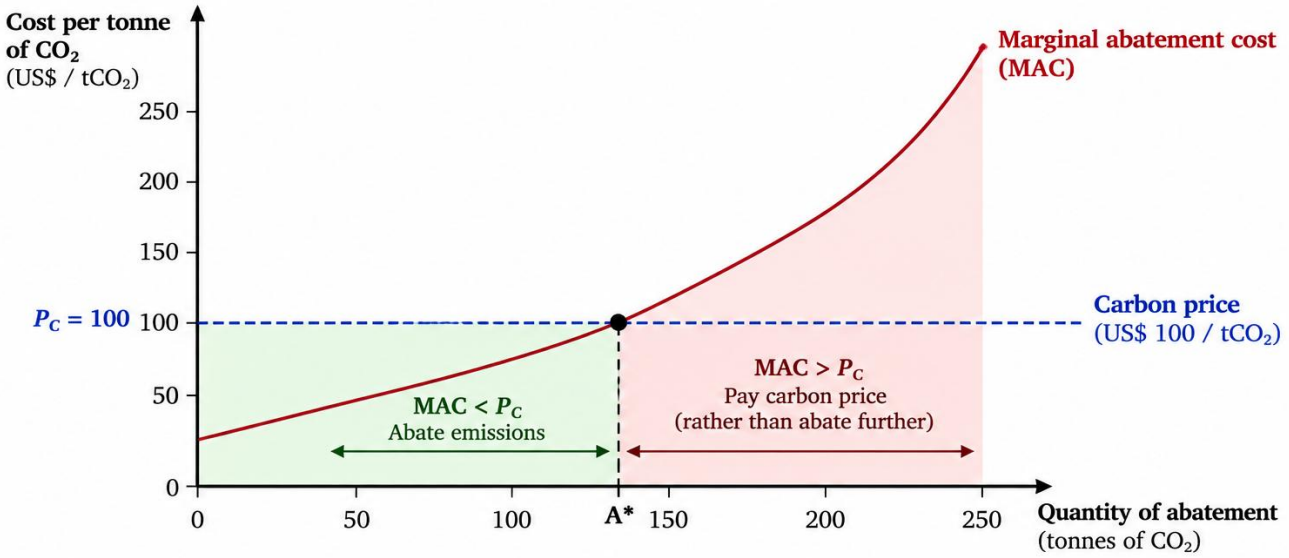
For Ship B:

$$MAC_B > P_C$$

Pay the carbon price rather than undertake that additional unit of abatement

2.4.2 Marginal Abatement Cost (MAC)

Graph 5 – Marginal Abatement Cost and Carbon Price



A cost-minimising firm abates up to the point where the marginal abatement cost equals the carbon price (MAC = P_C).

If MAC is below the carbon price, it is cheaper to abate. If MAC is above the carbon price, it is cheaper to pay the carbon price.

2.4.3 Cost-Effectiveness

- **Cost Effectiveness**

Achieving a given regulatory or environmental objective at the **lowest possible total cost**.

The cost-effective allocation occurs when:

$$MAC_A = MAC_B = \ldots = MAC_N$$

2.4.3 Cost-Effectiveness

- **Example**

$$MAC_A = \$40/tCO_2; MAC_B = \$150/tCO_2$$

$$P_C = \$100/tCO_2$$

We could shift one unit of abatement from B to A and save \$150 - \$40 = \$110

Therefore, if MAC differs across firms, the same total emission reduction can be achieved at a lower cost by shifting abatement from high-MAC to low-MAC firms.

Cost–effective allocation: $\boldsymbol{MAC_A = MAC_B}$

2.4.3 Cost-Effectiveness

	Economic Efficiency	Cost-Effectiveness
Main question	How much should we regulate?	How should we achieve the target?
Regulatory target	Target is to be determined	Target is already given
Decision rule	Regulate until $MB_R = MC_R$	Allocate abatement until MACs are equalised
Condition	$MB_R = MC_R$	$MAC_A = MAC_B = \cdots = MAC_N$
Objective	Maximise social welfare	Minimise total compliance cost
Example	Should CO_2 emissions be reduced by 20%, 40% or 60%?	If the target is 40%, which ships/firms should undertake the reductions?

2.4.4 Carbon Tax vs Emission Trading

CARBON TAX Fix PRICE → Quantity adjusts	EMISSIONS TRADING Fix QUANTITY → Price adjusts
Regulator sets: Carbon price	**Regulator sets:** Emission cap
Example: **US$100/tCO₂**	Example: **1 million tCO₂**
Firms compare **MAC with carbon tax** $MAC < P_C$: **Abate** $MAC > P_C$: **Pay tax**	Firms **buy/sell emission allowances** Low MAC: **Abate & sell allowances** High MAC: **Buy allowances**
Regulator determines price Market determines total emissions	Regulator determines total emissions Market determines price

Both systems are flexible and encourage cost-effective emission reduction

2.4.5 Which Regulatory Instrument should we choose?

- **Command and Control Regulation**
  - Minimum safety standards are essential
  - Consequences of non-compliance can be catastrophic
  - Certain behaviours are simply unacceptable
  - Monitoring outcomes is difficult

2.4.5 Which Regulatory Instrument should we choose?

- **Market-based instruments**
  - Firms have very different compliance costs
  - Many technological solutions are available
  - Regulators cannot easily identify the cheapest solution
  - Flexibility encourages innovation

2.5 Case Study: FuelEU Maritime

- **FuelEU combines a mandatory performance standard with flexible compliance mechanisms.**

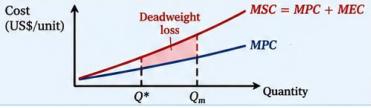
Mandatory Performance Standard	Flexible Compliance
Mandatory GHG-intensity limit	Firms choose how to reduce GHG intensity
Target becomes progressively stricter	Choice of fuels, technologies and operations
Compliance is monitored and verified	Banking of compliance surplus
Non-compliance can result in penalties	Borrowing from future compliance balance
Regulator specifies the outcome, not one technology	Pooling compliance balances across ships

Summary

Part II Summary
How Much and How Should We Regulate?

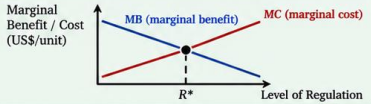
1 Market failure exists

Markets can produce inefficient outcomes when external costs or benefits are not reflected in private decisions.



2 Determine the appropriate level of regulation

Use marginal analysis to set the level of regulation where the marginal benefit equals the marginal cost of regulation.



3 Internalise the externality

Make the decision-maker face the external cost (or benefit) so that private costs reflect social costs.

$$MPC + MEC = MSC$$

- ✓ Pigouvian tax/charge
- ✓ Tradable permits
- ✓ Liability rules
- ✓ Performance standards

4 Choose a regulatory instrument

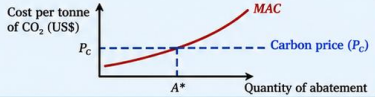
Select between command-and-control and market-based instruments, considering effectiveness, cost-effectiveness, flexibility, enforceability and information requirements.

Command-and-control
- Mandatory standards
- Technology requirements
- Prescriptive rules

Market-based
- Taxes / levies
- Emissions trading
- Performance-based regulation (e.g. FuelEU)

5 Allow low-cost abatement where appropriate

Firms abate up to the point where their marginal abatement cost equals the carbon price (or permit price).



6 Achieve the regulatory objective cost-effectively

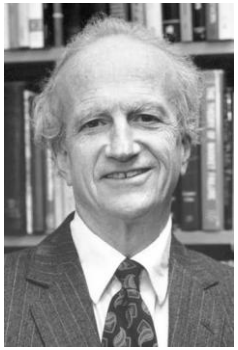
For a given target, abatement should be allocated across firms so that marginal abatement costs are equal (avoiding higher costs by shifting abatement from high-MAC to low-MAC firms).

$$MAC_A = MAC_B = \cdots = MAC_N$$

Same environmental outcome at the lowest possible total cost.

Good regulation is not simply the strictest regulation.
It seeks to achieve socially desirable outcomes at the lowest possible social cost.

3. How do firms and markets respond once regulation is imposed?



Gary Becker
Pioneer of Economics of Deterrence

By the end of Part III, students should understand how regulation affects costs, prices and firm behaviour, including who ultimately bears regulatory costs and how firms respond to compliance and enforcement incentives.

3.1 Regulation Creates a Cost Shock

- **New Regulation**

New regulations incur costs to shipowners such as carbon charges, cleaner fuels, equipment retrofits, training, and monitoring and reporting.

Shipowners can respond through (1) Absorb, (2) Pass on, (3) Adapt, (4) Innovate, and (5) Exit.

3.1 Regulation Creates a Cost Shock

- **New Regulation**

Response	What does the shipowner do?	Economic effect	Maritime example
1. Absorb	Pays the additional regulatory cost without changing the price charged	Profit margin falls	Carbon cost rises by $100/container, but freight rate is unchanged
2. Pass on	Transfers some or all of the cost to customers through higher prices	Freight rate rises	$100 additional cost → freight rate increases by $70
3. Adapt	Changes existing operations to reduce the cost of compliance	Operational behaviour changes	Slow steaming, route optimisation, improved vessel deployment
4. Innovate	Invests in new technology or capabilities to reduce compliance cost	Technology/capital changes	Alternative-fuel vessel, wind assistance, energy-saving technology
5. Exit	Stops serving the market/activity because compliance is no longer profitable	Supply/market structure changes	Older vessel scrapped; operator withdraws from a regulated route

3.1 Regulation Creates a Cost Shock

- **New Regulation**

Does the shipowner necessarily bear the entire regulatory cost?

3.2 Regulatory Incidence: Who really pays?

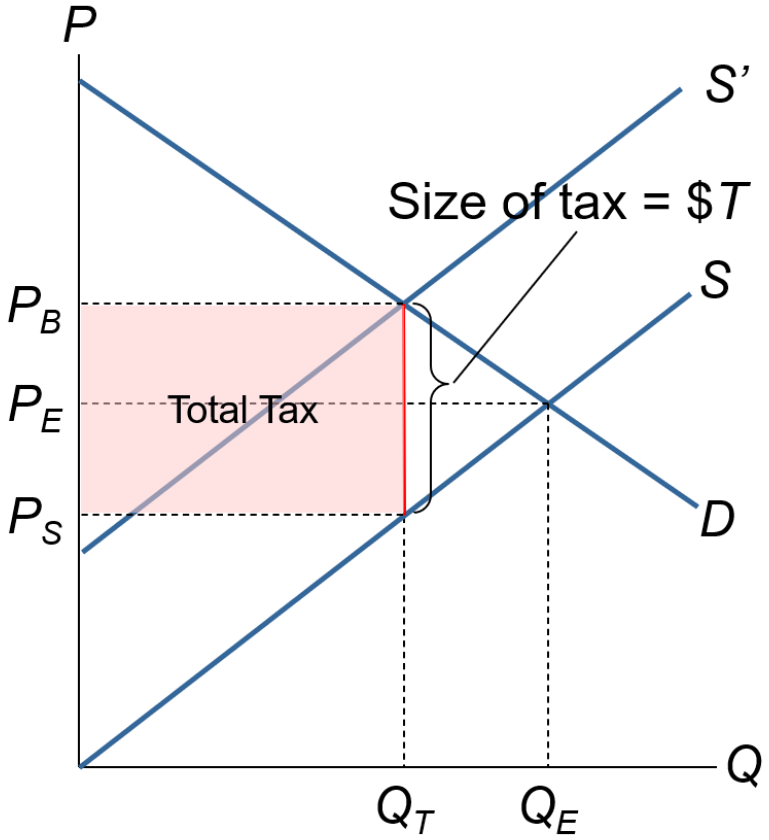
- **Regulatory Incidence**

Regulatory incidence refers to the manner in which the economic burden of a regulation is shared among participants in a market

Statutory incidence: who is legally responsible for the regulatory cost.

Economic incidence: who ultimately bears the cost.

3.3 Regulatory Costs: The Case of a Per-Unit Tax



$$\textbf{Following a regulation: } S \rightarrow S'; Q_E \rightarrow Q_T$$

$$\textbf{Price paid by buyers} = P_B$$

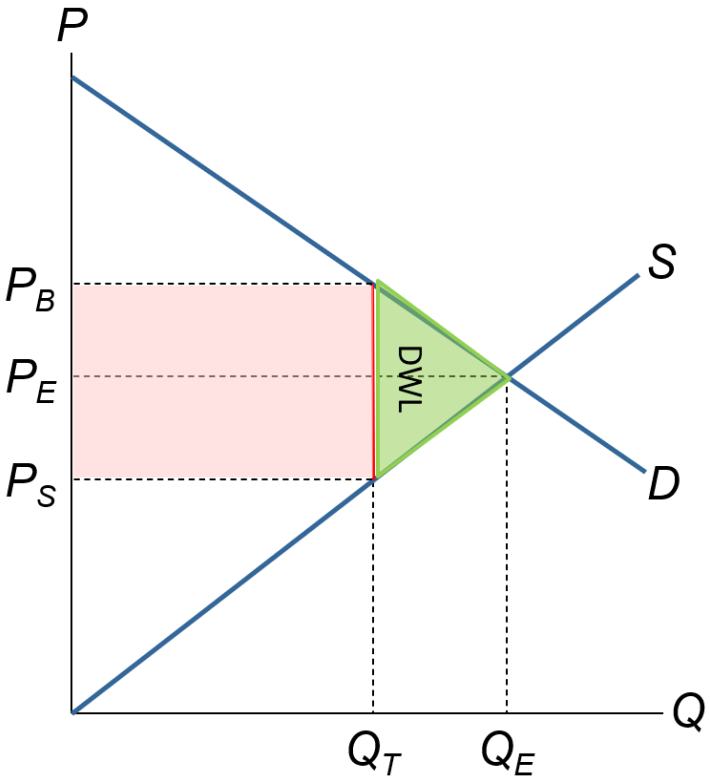
$$\textbf{Price received by sellers} = P_S$$

$$\textbf{Original Equilibrium Price} = P_E$$

$$\textbf{Total Tax Wedge} = T = P_B - P_S$$

$$\textbf{Total Tax} = T \times Q_T$$

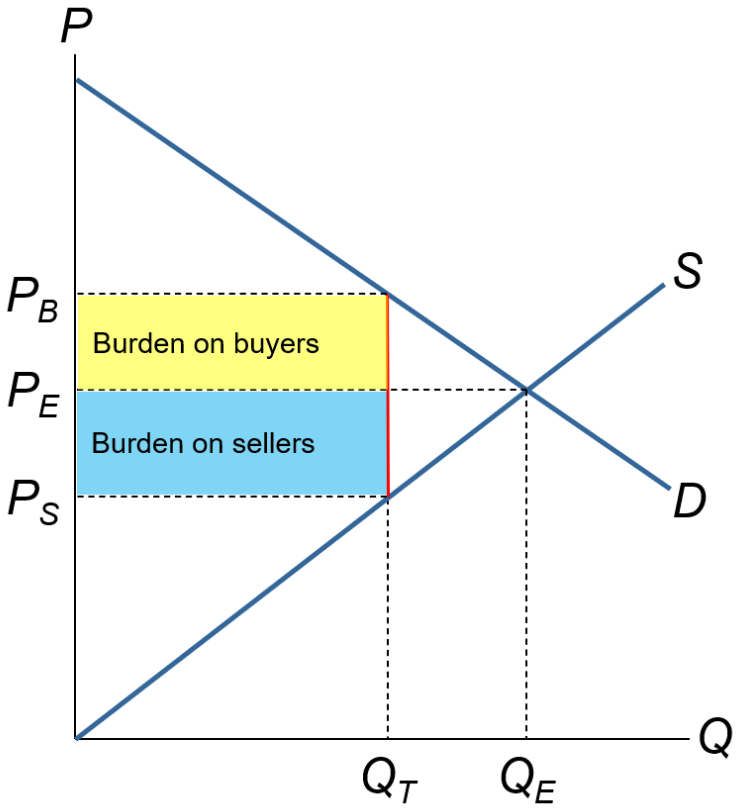
3.3 Regulatory Costs: The Case of a Per-Unit Tax



$$\textbf{Deadweight Loss } (DWL) = \frac{1}{2} \times T \times (Q_E - Q_T)$$

Deadweight loss (DWL) is the loss of total economic surplus caused when a tax or regulation prevents mutually beneficial transactions from taking place.

3.3 Regulatory Costs: The Case of a Per-Unit Tax



$$Buyer\ burden\ per\ unit = P_B - P_E$$

$$\textbf{Buyer burden} = (P_B - P_E) \times Q_T$$

$$Seller\ burden\ per\ unit = P_E - P_S$$

$$\textbf{Seller burden} = (P_E - P_S) \times Q_T$$

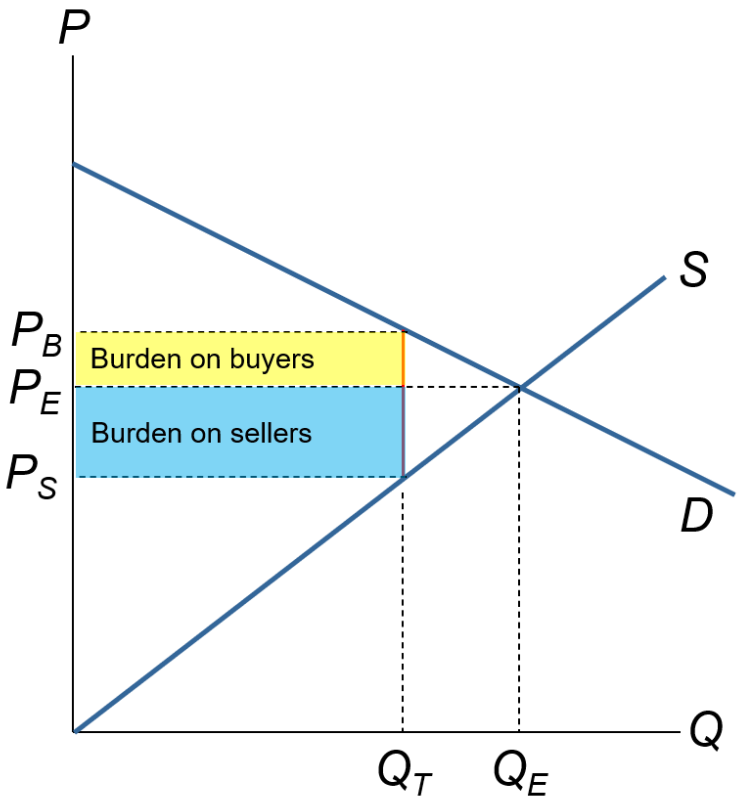
$$\textbf{Total tax} = \textbf{Buyer burden} + \textbf{Seller burden}$$

3.4 Effect of Price Elasticity on share of burden

- **Key Theory**

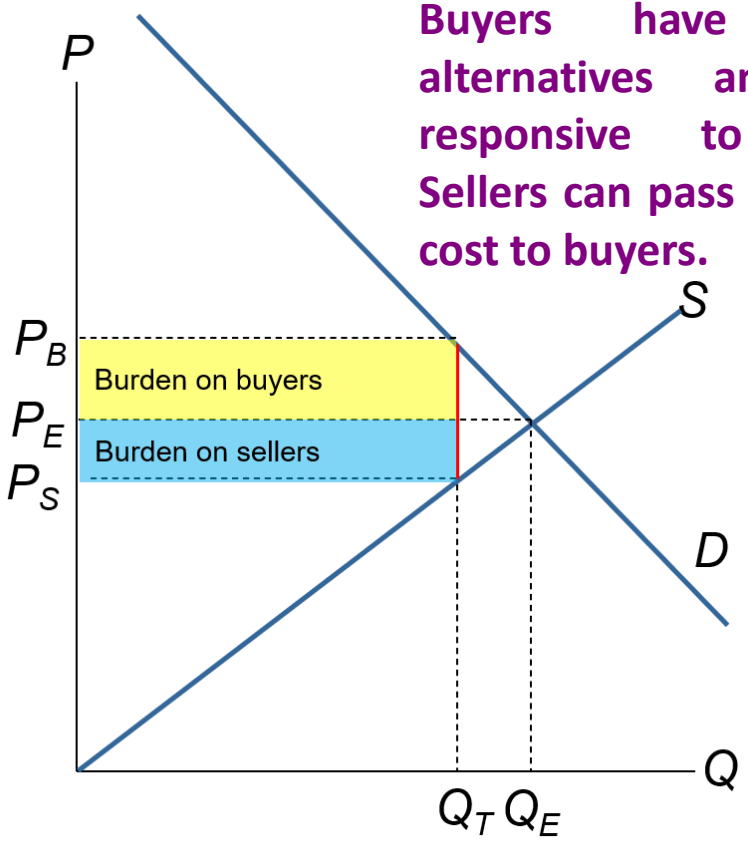
The less elastic side of the market bears more of the burden

3.4 Effect of Price Elasticity on share of burden



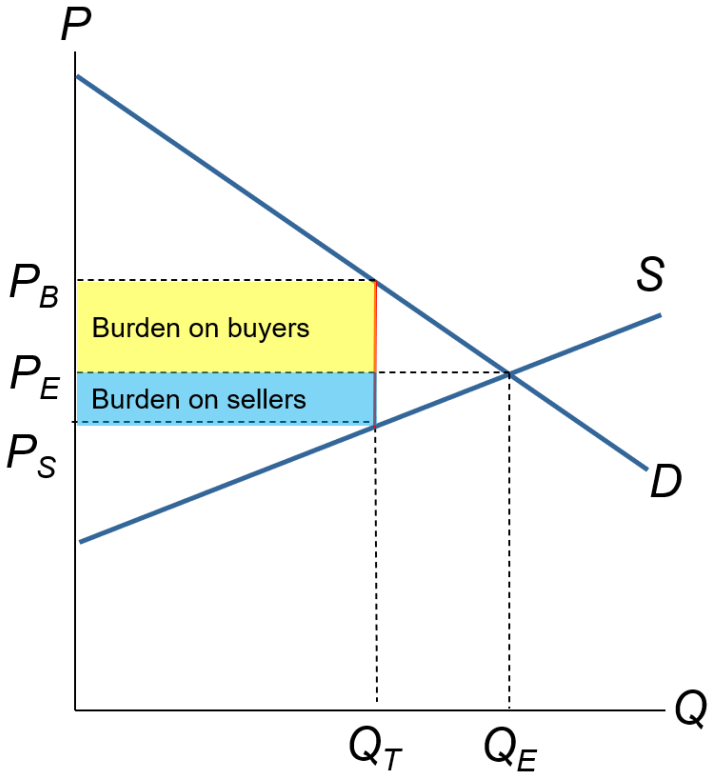
Elastic Demand

Buyers have fewer alternatives and less responsive to price. Sellers can pass on more cost to buyers.



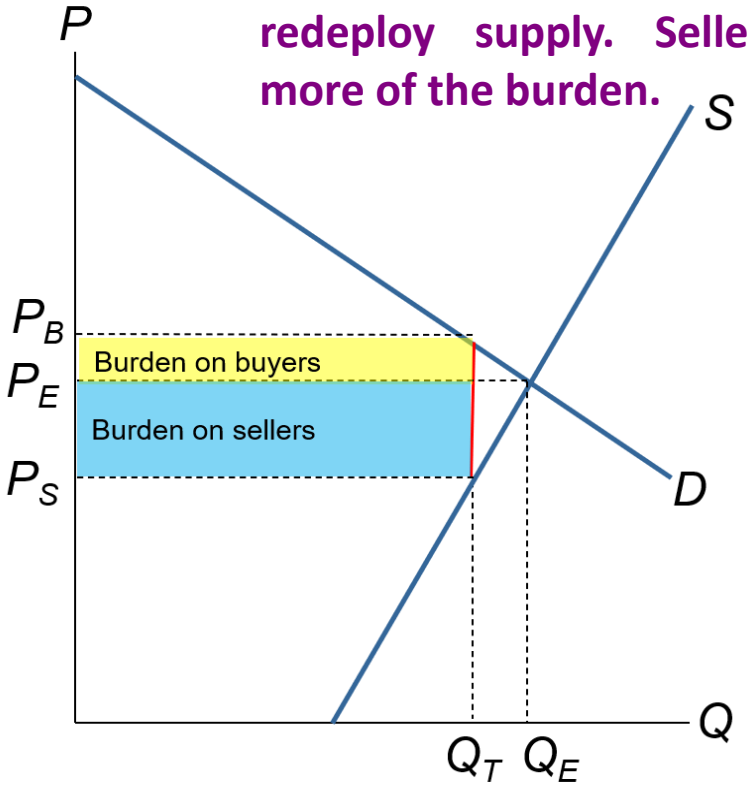
Inelastic Demand

3.4 Effect of Price Elasticity on share of burden



Elastic Supply

Sellers cannot reduce or redeploy supply. Sellers bear more of the burden.



Inelastic Supply

3.5 Economics of Deterrence

- **Economics of Deterrence**

If complying with a regulation is expensive, will every firm automatically comply?

3.5 Economics of Deterrence

- **Economics of Deterrence**

A firm has an incentive to violate a regulation when the expected benefit of non-compliance exceeds its expected cost.

3.5 Economics of Deterrence

- **Economics of Deterrence**

$$E(Penalty) = p \times F$$

Where:

p = probability of detection

F = fine / penalty if caught

3.5 Economics of Deterrence

- **Economics of Deterrence**

Scenario	Compliance cost	F	p	$E(Penalty)$	Description
A	100,000	500,000	5%	25,000	$E(Penalty)$ < Compliance Cost **Non-compliance is attractive**.
B	100,000	500,000	50%	250,000	$E(Penalty)$ > Compliance Cost **Compliance becomes economically attractive**.

Deterrence depends on both the probability of detection and the size of the penalty.

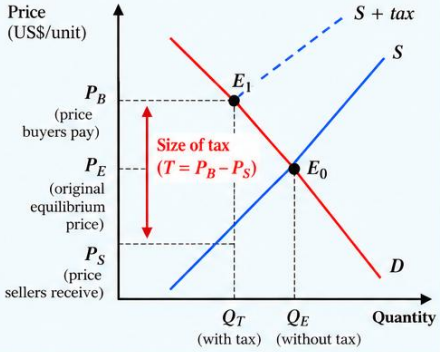
Effective Regulation $=$ Rules $+$ Monitoring $+$ Enforcement

Summary

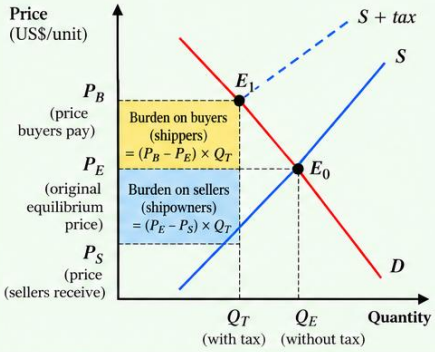
Part III | What Happens When We Regulate?

A tax creates a wedge, redistributes surplus, and reduces quantity

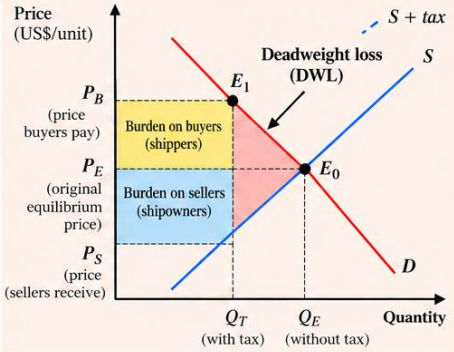
1 A tax creates a wedge



2 The tax burden is shared



3 The tax creates a deadweight loss



Key effects

- The tax shifts the supply curve upward by the size of the tax (T).
- Buyers pay a higher price (P_B).
- Sellers receive a lower price (P_S).
- The tax wedge is the vertical distance between P_B and P_S ($T = P_B - P_S$).
- Quantity traded falls from Q_E to Q_T.

Incidence depends on elasticity

- Total tax revenue collected equals $T \times Q_T$ and is the sum of the burdens on buyers and sellers.
- Burden on buyers = $(P_B - P_E) \times Q_T$.
- Burden on sellers = $(P_E - P_S) \times Q_T$.
- The less elastic side of the market bears a larger share of the tax.
- If demand is inelastic → shippers bear more.
- If supply is inelastic → shipowners bear more.

Deadweight loss

- The tax reduces quantity below the efficient level ($Q_T < Q_E$).
- The lost mutually beneficial transactions create a deadweight loss (triangle).
- DWL = ½ × $(P_B - P_S) \times (Q_E - Q_T)$ = ½ × $T \times (Q_E - Q_T)$.
- Tax revenue (rectangles) is a transfer from buyers and sellers to the government, not a loss of welfare.
- Deadweight loss is a net loss of total surplus.

4 How do firms respond?

Firms can adjust in different ways, depending on costs, market conditions and regulation.

- **Absorb** (lower profits)
- **Pass on** (higher freight rates)
- **Adapt** (improve efficiency)
- **Innovate** (cleaner technology)
- **Exit or redeploy** (shift to other trades)

Regulation changes incentives, prices and quantities — and its impacts depend on market elasticities and firm responses.

Reference

- Mankiw, N. G. (2027). *Principles of Microeconomics* (11th ed). South-Western Cengage Learning. [Chapters 7–8]

